Nikita, Shams, Michal

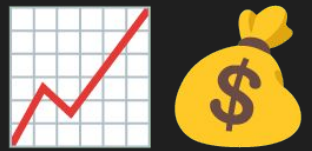
COOP PANK

Business Problem

Coop Bank wants to **build a portfolio**:

pick 5 stocks out of 10

return is high and risk is low, with sector limits.



Why is the problem difficult?

The number of possible portfolios explodes.

5 of 10 stocks -> **252 options**

10 of 100 stocks -> **17 trillion**

Real market like

NYSE -> **astronomically more.**

Because risk comes from every pair of stocks,

This is a known **NP-hard problem** (cardinality-constrained portfolio optimisation).

How many qubits are needed to solve full problem?

One qubit per stock:

1 = buy, 0 = don’t buy.

Our toy problem uses 10 qubits. The full problem needs N qubits: about 100 for a mid-size universe, around 2,400 for all of NYSE. Plus a few extra «slack» qubits per sector for the «at most 2» rule. With unequal weights it becomes N × (bits per weight).

Scaling

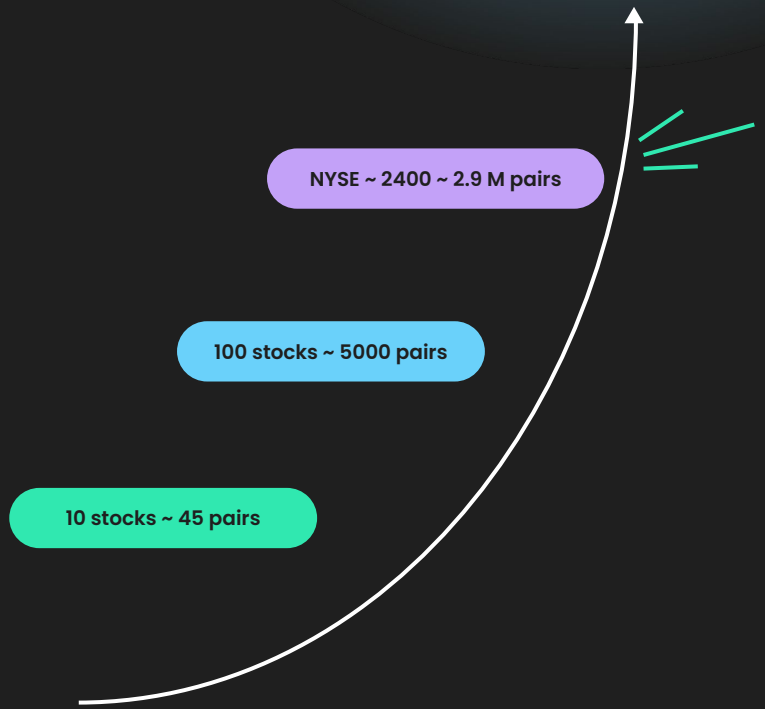
The number of **qubits grows linearly** with N,

but the **circuit grows quadratically**: every pair of stocks has a correlation term, and the «exactly K stocks» rule also links every pair.

N(N–1)/2 two-qubit interactions per QAOA

layer: 45 for our 10 stocks,

about 5,000 for 100 stocks, and millions for NYSE.

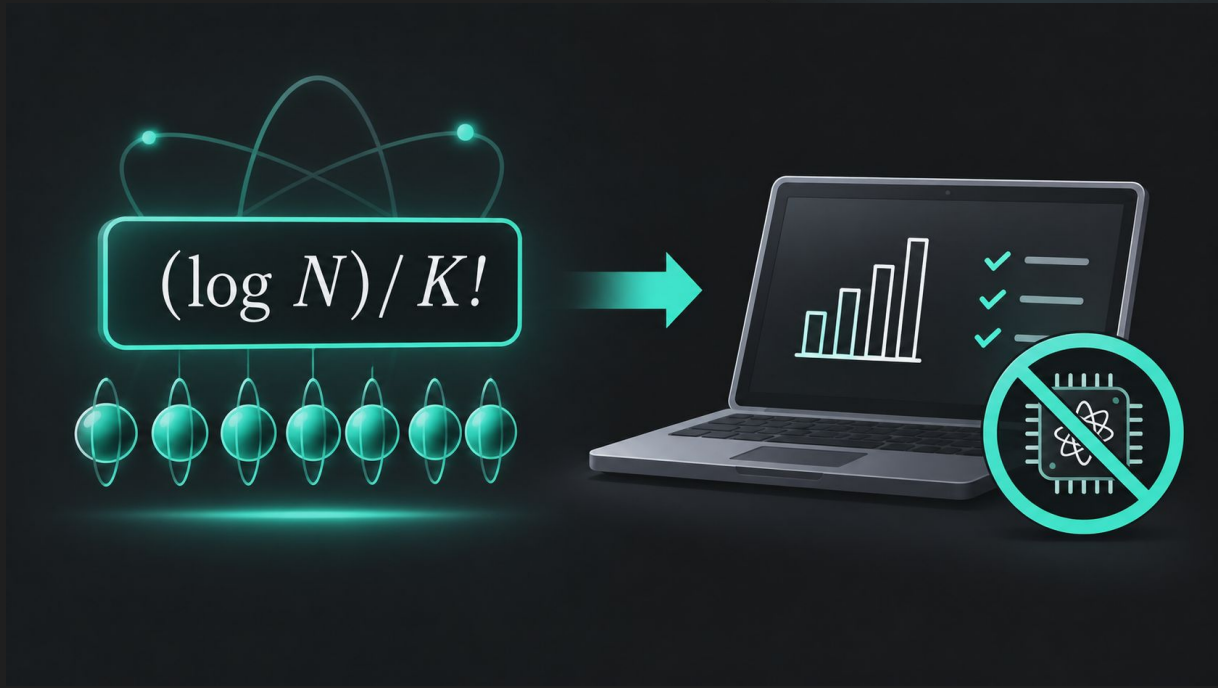


Scaling

Linear growth is the best solution we have found.

Another way would be to use **(logN)/K!** qubits. This way we encode each possible portfolio as a **unique identifier**.

Problem: all the weight and constraint calculations would happen on a classical computer. **No need for a quantum one.**



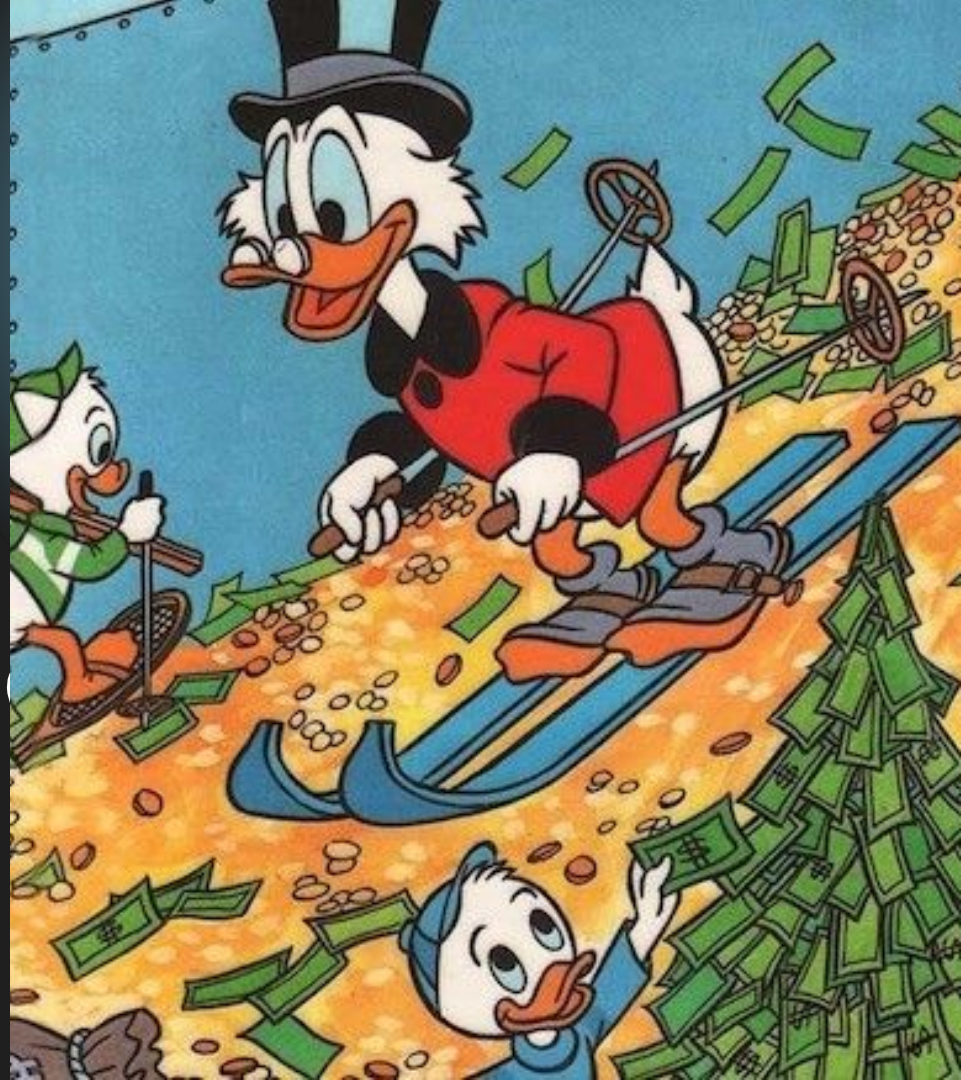
Results

We have successfully managed to find an optimal solution to choose **5 stocks out of the given 10** taking into account all of the following data:

1. **Annual return of each stock**
2. **Annual volatility of each stock**
3. **Correlation between stocks.**

The solution also includes the following constraints:

1. **Choose exactly 5 stocks**
2. **No more than 2 stocks from financial and consumer groups.**



Results

We choose the **5 stocks** that maximise expected return minus a risk penalty:

$R^T x - \alpha \cdot x^T \sum x$

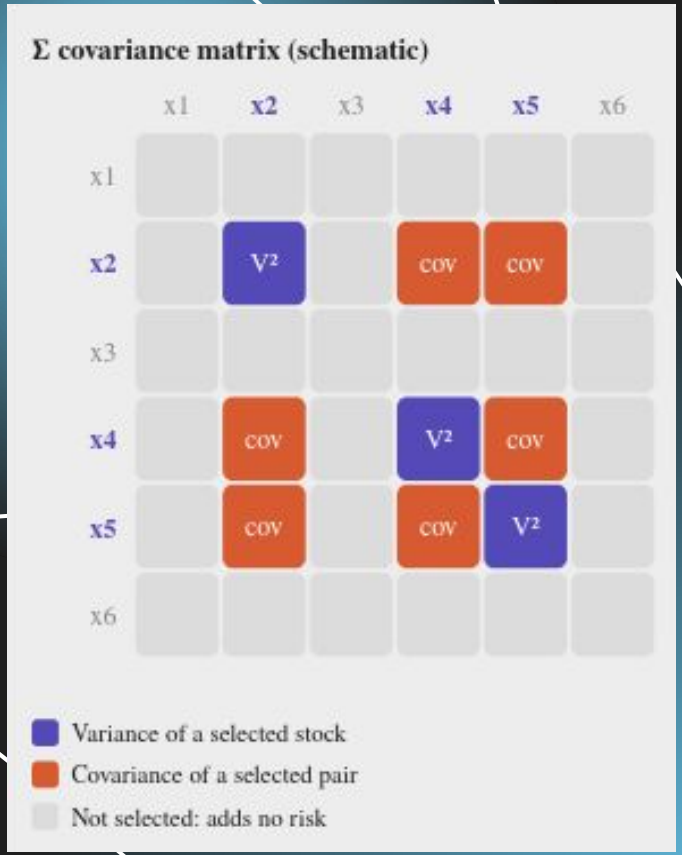
x – 1 if a stock is selected, otherwise 0

R – annual returns of the stocks

∑ – covariance matrix: variance V^2 on the diagonal, how two stocks move together everywhere else

α – risk parameter (0, 2 or 5)

Constraints: exactly 5 stocks, at most 2 per sector.



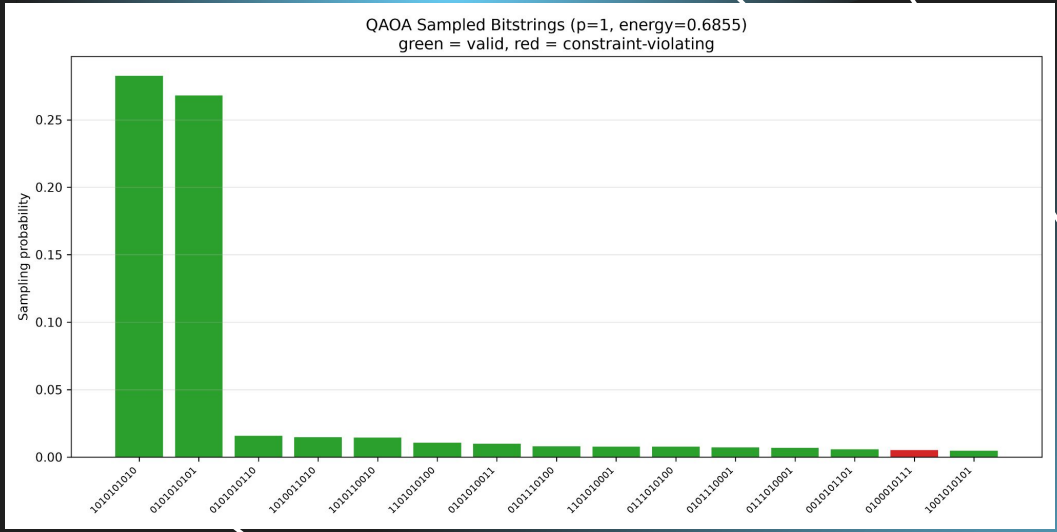
Add

Correlation changes which stocks are attractive as a group: the optimizer can favor combinations whose movements offset each other, because their covariance lowers total portfolio risk—not just stocks that look good individually.

Results

Based on the calculations, we have concluded that the best stock to buy at **layer 1 (p = 1)** are:

1. **LHV, Artea, Apranga, Ignitis and Tallink**
2. **Coop Pank, Tallinna Kaubamaja, Pieno žvaigždės, Tallinna Vesi, Tallinna Sadam**



Results

ibm_chicago



Results

11.66%	10.95%	1.06
Return	Volatility	Sharpe ratio

Tallinn — 3 of 15 selected

LHV1T q0	CPA1T q1	TAL1T q2	**TSM1T q3**	**TKM1T q4**	TVE1T q5	MRK1T q6	PRF1T q7	PKG1T q8
SKN1T q9	HAE1T q10	**ARC1T q11**	EFT1T q12	HPR1T q13	INF1T q14			

Riga — 0 of 5 selected

SAF1R q15	BAL1R q16	DGR1R q17	ELEVR q18	IDX1R q19

Vilnius — 2 of 13 selected

IGN1L q20	ROE1L q21	PZV1L q22	**APG1L q23**	TEL1L q24	AKO1L q25	**GRG1L q26**	INL1L q27	VLP1L q28
ZMP1L q29	AUG1L q30	KNE1L q31	KNR1L q32					

Lessons learned



Quantum solution is not always the best one

For 10 stocks, classical brute force finds the exact best portfolio in milliseconds. QAOA only makes sense at scale, and today's noisy hardware isn't there yet.

Build the rules into the circuit

Penalties for «exactly 5 stocks» made QAOA waste effort on invalid portfolios. A Dicke start state with an XY mixer keeps every sample a valid 5-stock portfolio: 252 options instead of 1,024.

Valid != optimal

QAOA always returned valid portfolios, but rarely the best one, because the top portfolios have very similar costs. Shallow circuits can't tell them apart. Choosing the best of all samples works better than taking the most frequent one.

Next steps

Sell for a billion euros;)

Anyone interested?

Thank you!

